

Principles of Communication Systems

Quiz #1

Instructor: Ranjitha Prasad

Duration: 75 minutes

Please provide your answers clearly, with all the steps outlined. Some of the trigonometric identities required are as follows:

- $\sin(2\theta) = 2\sin(\theta)\cos(\theta)$
- $\cos(2\theta) = \cos^2(\theta) - \sin^2(\theta)$
- $\cos(\theta_1 - \theta_2) = \cos(\theta_1)\cos(\theta_2) + \sin(\theta_1)\sin(\theta_2)$
- Signal $I[a, b](t)$ is 1 between a and b and zero otherwise.

Problem 1 Determine if the system with the input-output relationship given by

$$y(t) = \int_{t-10}^t \cos(\tau)x(\tau)d\tau,$$

is linear and time invariant (LTI). Provide the steps clearly. .. (2)

Problem 3 The inner product (correlation) of two real-valued passband signals $u_p(t)$ and $v_p(t)$ is defined as,

$$\langle u_p, v_p \rangle = \int_{-\infty}^{\infty} u_p(t)v_p(t)dt.$$

Answer the following questions:

- (a) Given the rectangular form as $u_p(t) = u_c(t)\cos(2\pi f_c t) - u_s(t)\sin(2\pi f_c t)$ (similar for $v_p(t)$). Using the rectangular representation of the passband signals at f_c , obtain the expression for $\langle u_p, v_p \rangle$ in terms of $\langle u_c, v_c \rangle$ and $\langle u_s, v_s \rangle$..(2)
- (b) Consider two passband signals $s_{p1}(t) = \sqrt{\frac{2}{T}}(u(t) - u(t-T))\cos(2\pi f_c t)$ and $s_{p2}(t) = \sqrt{\frac{2}{T}}(u(t) - u(t-T))\sin(2\pi f_c t)$, $f_c \gg \frac{1}{T}$. Let $s_1(t)$ and $s_2(t)$ be the complex baseband representations of $s_{p1}(t)$ and $s_{p2}(t)$. Compute $\langle s_1, s_2 \rangle$ and $\langle s_{p1}, s_{p2} \rangle$ and conclude if (a) orthogonality in baseband implies orthogonality in passband? (b) orthogonality in passband implies orthogonality in baseband? ..(3)

Problem 3 A carrier signal $c(t) = A_c \cos(2\pi f_c t)$ is modulated by a single tone signal $m(t) = A_m \cos(2\pi f_m t)$ to obtain .. (3)

$$s(t) = [A_c + A_m \cos(2\pi f_m t)]\cos(2\pi f_c t).$$

Let $A_c = 10$, $A_m = 2$, $f_c = 1000\text{Hz}$ and $f_m = 50\text{Hz}$. Answer the following:

- (a) Rewrite $s(t)$ to obtain the sidebands of $s(t)$ in time domain.
- (b) Sketch $s(t)$. Depict the message signal, carrier waveform and carefully mark the amplitude levels.
- (c) Sketch the frequency spectra: $|M(f)|, |C(f)|, |S(f)|$ where $m(t) \leftrightarrow M(f)$, $c(t) \leftrightarrow C(f)$ and $s(t) \leftrightarrow S(f)$.
- (d) Compute the modulation index of $s(t)$.

Problem 4 The passband signal $v_p(t) = I[-2, 2](t)\cos 100\pi t$ is passed through the passband filter $h_p(t) = I[0, 3](t)\sin 100\pi t$. .. (5)

1. Find an explicit time domain expression for the filter output $y_p(t)$.
2. Find the output when $y_p(t)\cos 101\pi t$ is passed through a lowpass filter.

Problem 5 A message to be transmitted using AM is given by $m(t) = 3\cos 2\pi t + 4\sin 6\pi t$ where the unit of time is milliseconds. It is to be sent using a carrier frequency of 600 KHz...(5)

1. For a modulation index of 50 %, write an explicit time domain expression for the AM signal.
2. What is the power efficiency of the AM signal?
3. Sketch the magnitude spectrum for the AM signal, again clearly specifying the units used on the frequency axis.
4. The AM signal is to be detected using an envelope detector, with $R = 50$ ohms. What is a good range of choices for the capacitance C ?